A Unified Multi-Modal Transit Fare Comparison Framework and Explainable Journey Optimization System for Dynamic Urban Commutes

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***Abstract*— Rapid urbanization and fragmented transit ecosystems in metropolitan cities like Bengaluru create severe information asymmetry, sub-optimal commuter routing, and unpredictable travel expenses. Commuters routinely navigate disparate public transit networks (BMTC bus routes, Namma Metro rail) and competing private ride-hailing services (Ola, Uber, Rapido, Namma Yatri), alongside personal vehicles. Existing navigation platforms either omit real-time fare breakdowns, lack multi-modal interchange feasibility, or treat routing as a black box without transparent trade-off justification. In this paper, we propose the Unified Transit Routing System (UTRS), an end-to-end framework that synchronizes public transit schedules, calibrated ride-hailing rate cards, dynamic surge heuristics, personal vehicle lifecycle costs, and real-time weather constraints into a Pareto-optimal multi-criteria journey optimizer. Our engine introduces a Weather-Adaptive Multi-Criteria Cost-Time-Fatigue objective function and a novel Marginal Cost of Time Saved (MCTS) metric within an Explainable AI (XAI) reasoning layer to justify trade-offs in natural language. Experimental benchmarks across high-density urban corridors demonstrate that UTRS computes comprehensive multi-modal routes in <120 ms, identifies cost reductions of up to 58% compared to direct ride-hailing, and improves transit interchange efficiency under adverse weather.**

***Keywords*— Multi-Modal Transit Optimization, Fare Estimation, Algorithmic Surge Pricing, Explainable AI (XAI), Intelligent Transportation Systems (ITS), GTFS Routing, Dynamic Urban Mobility.**

I. INTRODUCTION

A. Background and Motivation— Metropolitan hubs in rapidly growing economies present a uniquely complex and fragmented mobility landscape. In Bengaluru, India—frequently cited as having among the highest traffic congestion indices globally—commuters are tasked with navigating multiple independent transit networks daily:

1) *Public Transit Infrastructure:* The Bangalore Metropolitan Transport Corporation (BMTC) bus network spanning over 6,000 scheduled routes and 2,000+ transit stops, complemented by the Namma Metro rail network operated by the Bangalore Metro Rail Corporation Limited (BMRCL).

2) *On-Demand Private Aggregators:* Commercial ride-hailing services (Uber, Ola, Rapido) offering two-wheeler bikes, three-wheeler auto-rickshaws, and multi-tier cabs, along with open-protocol decentralized alternatives (Namma Yatri).

3) *Personal Vehicles:* Two-wheelers and four-wheelers whose true commute costs vary dynamically based on certified fuel efficiency, daily retail petrol/diesel prices, and distance-proportional maintenance amortization.

B. The Commuter's Dilemma— Despite widespread smartphone penetration, commuters suffer from severe information asymmetry. Commercial mapping applications (e.g., Google Maps) provide robust road network routing but omit synchronized cross-provider fare comparisons, real-time dynamic surge estimation, night-time surcharges, platform-specific booking fees, and multi-modal transfer feasibility under adverse meteorological conditions. Consequently, commuters are forced to manually switch between multiple proprietary mobile applications to cross-examine prices and arrival estimates, leading to suboptimal travel choices, elevated personal expenses, and urban congestion.

C. Research Contributions— To resolve this fragmentation, this paper presents the Unified Transit Routing System (UTRS). The primary contributions of this research are as follows:

• **Mathematical Multi-Modal Rate Card Formalization:** Reverse-engineering and encoding algorithmic rate cards across four ride-hailing platforms, BMTC stage-fare matrices, Metro station distance slabs, and personal vehicle depreciation curves into a synchronized computational pipeline.

• **Weather-Adaptive Multi-Criteria Optimization:** Formulating a multi-objective cost-time-fatigue routing model that dynamically adapts pedestrian walking speeds and transfer delay buffers based on precipitation conditions.

• **Explainable AI (XAI) Commuter Decision Layer:** Introducing the Marginal Cost of Time Saved (MCTS) metric coupled with Natural Language Generation (NLG) templates to provide clear, transparent justifications for transit recommendations.

• **End-to-End System Deployment:** Implementing an asynchronous, high-throughput REST backend (FastAPI) paired with an interactive modern client (React/Vite and Flutter) achieving sub-120 ms latency across complex urban corridors.

II. RELATED WORK AND LITERATURE REVIEW

Urban transit journey planning has traditionally relied on graph-theoretic shortest path algorithms, such as Dijkstra's algorithm, Contraction Hierarchies, Round-Based Public Transit Routing (RAPTOR), and the Connection Scan Algorithm (CSA). While computationally efficient on timetable-based transit networks, conventional algorithms evaluate single-objective functions (typically shortest physical distance or travel time) and fail to incorporate multi-provider monetary fare structures, dynamic surge variables, and micro-mobility links.

Recent research in on-demand mobility platforms has investigated algorithmic dynamic pricing mechanisms driven by spatial-temporal demand-supply imbalances. However, existing literature primarily approaches dynamic pricing from the aggregator's revenue-maximization perspective rather than the commuter's multi-modal cost-optimization perspective. Furthermore, while Explainable Artificial Intelligence (XAI) has made substantial inroads into healthcare and finance, its application within Intelligent Transportation Systems (ITS) has remained largely confined to traffic signal automation, leaving commuter-facing decision support largely unexplained.

III. SYSTEM ARCHITECTURE & DATA ENGINEERING

The UTRS framework is organized as a modular, five-tier decoupled architecture designed for sub-second query processing, high fault tolerance, and multi-platform compatibility.

A. GTFS Public Transit Ingestion & Graph Modeling

The BMTC bus transit network is ingested from General Transit Feed Specification (GTFS) feeds and normalized into a directed weighted graph G_bus = (V_bus, E_bus). Vertices V_bus represent verified bus stop coordinates, and edges E_bus encode directional road links weighted by scheduled inter-stop running time and Euclidean distance. The Namma Metro rail topology is structured as a dual-line network (Purple and Green lines) supporting bidirectional Breadth-First Search (BFS) to identify shortest topological paths and interchange nodes (e.g., Kempegowda Majestic Interchange).

B. Multi-Provider Ride-Hailing Rate Card Modeling

Commercial ride-hailing services implement distinct, proprietary pricing functions. UTRS formalizes these into unified mathematical engines for provider p in {Uber, Ola, Rapido, Namma Yatri} and vehicle category v:

$$C_{p,v}(d, t) = max(C_{min}, [C_{base} + f_{dist}(d) + f_{time}(t) + C_{pickup} + C_{booking}] \cdot S(t_{IST}) \cdot N(t_{IST})) + C_{toll} \quad (1)$$

Where d represents route distance (km), t is estimated in-vehicle duration (min), S(t_IST) is the time-of-day surge multiplier (ranging from 1.0 to 1.5 during peak hours 08:00–10:00 and 17:00–21:00 IST), N(t_IST) is the nocturnal surcharge multiplier (1.5 for Autos and 1.1 for Cabs between 22:00 and 05:00 IST), and C_toll represents airport expressway corridor levies (■120).

C. Personal Vehicle Operating Cost Model

For personal two-wheelers and four-wheelers, total commute expenditure is modeled by integrating real-time fuel market pricing (P_fuel), certified vehicle fuel economy (M_km/L), and per-kilometer component depreciation (K_maint):

$$C_{vehicle}(d) = d \cdot [(P_{fuel} / M_{km/L}) + K_{maint}] \quad (2)$$

IV. MATHEMATICAL FORMULATIONS & ROUTING OPTIMIZATION

A. Unified Multi-Criteria Cost-Time-Fatigue Objective Function

Rather than isolating cost or duration, UTRS formulates journey evaluation as a multi-criteria penalty minimization problem across all candidate paths P:

$$J(P) = T_{total}(P, W) + \alpha \cdot C_{total}(P) + \beta \cdot N_{transfers}(P) + P_{walk}(d_{walk}, W) \quad (3)$$

Where T_total(P, W) represents total travel duration in minutes under weather state W, C_total(P) is total monetary fare in INR, α is the commuter's cost-sensitivity weighting parameter (α = 1.5 for budget mode, α = 0.1 for rapid transit mode), β is the transfer fatigue penalty constant (β = 10.0 min per modal switch), and P_walk represents non-linear penalties for walking distances exceeding 500 meters.

B. Weather-Adaptive Degradation Model

Adverse weather significantly impedes pedestrian speed and expands transfer friction at open-air bus interchanges. The weather degradation functions are defined as:

$$V_{walk}(W) = \{ 5.0 \text{ km/h (Clear)}, 4.25 \text{ km/h (Light Rain)}, 3.25 \text{ km/h (Heavy Rain)} \} \quad (4)$$

$$\Delta T_{buffer}(W) = \{ 3 \text{ min (Clear)}, 6 \text{ min (Light Rain)}, 12 \text{ min (Heavy Rain)} \} \quad (5)$$

C. Multi-Modal Hub Discovery & Bridge Synthesis

Intermodal transfer nodes are resolved dynamically by querying public transit hubs within walking radius R_walk_max = 500m of the commuter's origin and destination coordinates. If R_walk exceeds 500m, the engine automatically synthesizes micro-mobility bridge legs (e.g., Rapido bike or Auto-rickshaw feeder) to connect the commuter seamlessly to the nearest rapid transit trunk line.

V. EXPLAINABLE AI (XAI) & COMMUTER DECISION SUPPORT

To overcome commuter distrust in automated recommendations, UTRS incorporates an Explainable AI (XAI) decision layer that evaluates Pareto-dominant trade-offs among the top three candidate routes.

A. Marginal Cost of Time Saved (MCTS)

When comparing a faster, more expensive option (J_fast) against an economical alternative (J_cheap), the system computes the exact economic value of time saved:

$$MCTS(J_{fast}, J_{cheap}) = [(C(J_{fast}) - C(J_{cheap})) / (T(J_{cheap}) - T(J_{fast}))] \cdot 60 \quad [₹ / hour] \quad (6)$$

B. Natural Language Explanation Generation

Based on calibrated MCTS thresholds, UTRS generates natural language justifications rendered directly on the client UI:

• **High Efficiency (MCTS < ₹100/hr):** 'Highly recommended: Namma Metro bypasses road congestion, saving 32 minutes for only ₹18 extra over bus transit.'

• **Value Choice (MCTS > ₹350/hr):** 'Recommended for budget: Direct BMTC bus saves ₹240 compared to ride-hailing, incurring only 12 minutes of additional transit time.'

• **Weather Contingency:** 'Weather Alert: Metro rail recommended over low-lying bus corridors to prevent flood-induced delays.'

VI. EXPERIMENTAL EVALUATION & PERFORMANCE BENCHMARKS

We evaluated the UTRS framework across three major benchmark commuter corridors in Bengaluru, representing diverse traffic, transit density, and distance profiles.

Corridor / Metric	Uber/Ola Cab	Namma Yatri Auto	BMTC Bus	Metro + Bus (UTRS)	Personal 2W
A: Silk Board → Indiranagar (Fare)	₹240 – ₹310	₹165	₹20	**₹38**	₹32.50
A: Silk Board → Indiranagar (Time)	42 min	48 min	65 min	**36 min**	38 min
B: Electronic City → Airport (Fare)	₹1,150 – ₹1,480	N/A	₹280 (V.Vajra)	**₹180**	₹145.00
B: Electronic City → Airport (Time)	85 min	N/A	110 min	**92 min**	90 min
C: Majestic → Whitefield (Fare)	₹460 – ₹580	₹310	₹25	**₹60 (Purple)**	₹62.00
C: Majestic → Whitefield (Time)	78 min	85 min	105 min	**44 min**	55 min

A. Cost and Time Analysis— As shown in Table I, multi-modal hybrid routes generated by UTRS consistently outperform direct ride-hailing on cost by 45% to 58%, while remaining competitive on travel time (and frequently faster during peak congestion hours via dedicated metro rail right-of-way).

B. Computational Latency— Benchmarking over 1,000 synthetic multi-modal coordinate queries yielded a mean end-to-end API execution latency of 98.6 ms, with graph traversal consuming 18.4 ms ± 2.1 ms and multi-provider fare matrix calculation consuming 12.1 ms ± 1.4 ms, validating real-time viability.

VII. CONCLUSION & FUTURE WORK

In this paper, we introduced UTRS, a comprehensive multi-modal transit comparison and journey planning framework addressing urban transit fragmentation in Bengaluru. By integrating calibrated ride-hailing pricing models, GTFS public transit graphs, weather-adaptive routing, and Explainable AI trade-off analytics, UTRS delivers transparent, cost-effective, and resilient transit recommendations.

Future research will focus on integrating live GPS telemetry streams from BMTC buses via vehicle tracking APIs, training LSTM/Transformer models for predictive surge pricing forecasts 30–60 minutes in advance, and extending the personal vehicle garage module to incorporate Electric Vehicle (EV) battery state-of-charge (SoC) and charging station routing.

REFERENCES

[1] J. Wardman, “Public transport values of time,” Transport Policy, vol. 11, no. 4, pp. 363–377, 2004.

[2] D. Delling, P. Sanders, D. Schultes, and D. Wagner, “Engineering route planning algorithms,” Algorithmics of Large and Complex Networks, Springer, pp. 117–139, 2009.

[3] R. Geisberger, P. Sanders, P. Schultes, and D. Delling, “Contraction hierarchies: Faster and simpler hierarchical routing in road networks,” in Experimental Algorithms, Springer, 2008, pp. 319–333.

[4] K. Cohen, M. Fishelson, and Y. Mansour, “Dynamic pricing in ride-hailing platforms,” in Proc. ACM Conf. Econ. Comput. (EC), 2022, pp. 112–134.

[5] G. Adomavicius and A. Tuzhilin, “Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions,” IEEE Trans. Knowl. Data Eng., vol. 17, no. 6, pp. 734–749, Jun. 2005.

[6] Bangalore Metropolitan Transport Corporation (BMTC), “Open Transit Data and GTFS Feed Documentation,” Bengaluru, India, 2024.

[7] Bangalore Metro Rail Corporation Limited (BMRCL), “Namma Metro Station Slabs and Fare Matrix Rules,” 2024.

[8] Beckn Protocol Foundation, “Open Mobility Specifications for Decentralized Transport Networks,” 2023.